

UNIVERSITY OF TECHNOLOGY - VNUHCM	HOMEWORK 5	Semester/Academic year	1 / 2026-2027
FACULTY OF COMPUTER SCIENCE AND ENGINEERING	Genetic Algorithms: Representation, Search, and Experimental Design	Due	12/10/2026 23:59
	CO3117 - Machine Learning	Format	Formative take-home

Purpose. Understand GA design decisions and how to evaluate search behaviour, without rehearsing the wording or sequence of the midterm.

- Complete the work by hand where calculation or derivation is requested. Show reasoning; unsupported final numbers receive limited credit.
- Use lecture notes and course resources. Do not use AI, online solution lookup, or classmates' answers for your first attempt. Later supportive AI use must follow the course AI_USE.md protocol.
- These activities are designed for concept transfer and assignment support. They intentionally do not reproduce the wording, ordering, numerical data, or question structure of the midterm.
- No official worked solutions will be released. Students are encouraged to self-check their handwriting with course materials, discussion, and AI assistants after completion.
- Submit one PDF in your course repository and preserve your first attempt. Corrections may be added later without replacing the original submission.

Activity 1 - Design a chromosome (1.5 pts)

You must select a subset of 12 candidate features. Propose a chromosome representation and explain how to decode it. Give one invalid or undesirable chromosome state and explain how your design handles it.

Activity 2 - Fitness under a trade-off (2.0 pts)

Define a fitness function that rewards validation accuracy but penalizes using many features. Evaluate your function for two hypothetical chromosomes: one using 3 features with accuracy 0.83 and one using 9 features with accuracy 0.88. Choose a penalty coefficient yourself and justify it.

Activity 3 - Search operators and constraints (2.0 pts)

- Start from the bit string 101001011100. Apply one mutation of your choice and one crossover with a second parent of your choice. Show the resulting offspring.
- Explain why the same operators may be inappropriate if the chromosome instead represents a permutation such as a TSP tour.

Activity 4 - Diagnose a GA run (2.0 pts)

A GA population reaches high average fitness quickly, but diversity collapses and the best fitness stops improving. Give two plausible causes and two interventions. For each intervention, explain the intended effect on exploration versus exploitation.

Activity 5 - Assignment bridge: bounded search experiment (2.5 pts)

- Use a GA for a small, bounded search related to your longitudinal assignment, such as feature selection or a restricted hyperparameter search. Define representation, fitness, population size, stopping rule, and the random-seed protocol.
- Run at least three independent seeds. Report best fitness by seed and one diversity or convergence indicator. Explain why a single successful run is not sufficient evidence.

Submission checklist

- ☐ Name and student ID on first page
- ☐ All calculations show intermediate steps
- ☐ Assignment-bridge activity uses your own dataset/use case
- ☐ First attempt preserved
- ☐ Any later AI/reference use disclosed according to course policy